

Basic Biosolids Calculations

This is intended as an aid for performing basic biosolids loading calculations for record keeping and reporting purposes.
This worksheet does not relieve the permittee from the responsibility for performing accurate calculations.

1. DETERMINE DRY TONS

Enter Percent Total Solids (%TS):	2.14
Enter Specific Bulk Density (lb/cubic yard) ¹ :	0
¹ (if quantity is in cubic yards)	
	Quantity Units
Enter the biosolids quantity and units:	60500 gallons
Results:	Dry Tons: 5.398899
	Dry Metric Tons (for EPA): 4.8977985696

2. DETERMINE DRY TONS PER ACRE

Enter Dry Tons:	5.398899
Enter Acres:	6.5
Results:	Dry Tons Per Acre: 0.8305998462
	Dry Metric Tons Per Hectare (for EPA): 1.8619498864

3. DETERMINE NUTRIENT LOADINGS

Enter Dry Tons Per Acre:	0.8305998462
Enter Percent Total Nitrogen (%TN) ² :	6.41
Enter Percent Total Phosphorus (%TP) ² :	5.76
Enter Percent Total Potassium (%TK) ² :	0.21

²Use this to convert from mg/kg to % if needed

mg/kg = 0%

Results:	lb/acre TN:	106.482900277
	lb/acre TP:	95.6851022769
	lb/acre TK:	3.4885193538
	kg/ha TN (for EPA):	119.351000873
	kg/ha TP (for EPA):	107.248325277
	kg/ha TK (for EPA):	3.9100951924

lb/acre PAN ³	70.988600185
lb/acre P ₂ O ₅	220.07573524

³ PAN based on 1.5 DEP factor. Calculations for PAN using EPA method are not provided.

4. DETERMINE METALS LOADINGS

Enter Dry Tons Per Acre: 0.8305998462

Enter Metals Concentrations (mg/kg):

		Loading Results:	
Metal	mg/kg	Metal	lb/acre
Arsenic	17.6	As	0.0292371146
Cadmium	4.1	Cd	0.0068109187
Copper	866	Cu	1.4385989335
Lead	21.9	Pb	0.0363802733
Mercury	4	Hg	0.0066447988
Molybdenum	26.7	Mo	0.0443540318
Nickel	30.2	Ni	0.0501682307
Selenium	5.7	Se	0.0094688382
Zinc	1630	Zn	2.7077554985

		kg/ha (for EPA)
Arsenic	17.6	0.0327703216
Cadmium	4.1	0.0076339954
Copper	866	1.6124487793
Lead	21.9	0.040776707
Mercury	4	0.0074478004
Molybdenum	26.7	0.0497140674
Nickel	30.2	0.0562308928
Selenium	5.7	0.0106131155
Zinc	1630	3.0349786493